Kiarash Rezaei

Doctoral Researcher | Chalmers University of Technology

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Research Interests

Trustworthy autonomous networks and O-RAN; uncertainty quantification, conformal prediction and risk control; reliable decision-making under distribution shift.

Education

PhD in Electrical Engineering (in progress) 2025 – Present
Chalmers University of Technology Gothenburg, Sweden
Department of Electrical Engineering. Supervisor: Prof. Paolo Monti; co-supervisor: Dr. Carlos Natalino.

MSc in Telecommunication Engineering 2021 – 2024
Politecnico di Milano Milan, Italy
Signal and Data Analysis track. Grade: 106/110.
Thesis: *Continuous IMU-MEMS Self-Calibration Process by Means of Tiny Neural Networks.*
Supervisor: Prof. Luca Barletta.

BSc in Computer Science 2014 – 2020
Kharazmi University Tehran, Iran
Department of Mathematics and Computer Science. Supervisor: Prof. Keivan Borna.
Thesis: *Automatic Architecture Design of CNNs using Genetic Algorithm and Reinforcement Learning (MetaQNN).*

Research Experience

Doctoral Researcher Feb 2025 – Present
Chalmers University of Technology Gothenburg, Sweden
Communication, Antennas and Optical Networks (CAOS) Division, Department of Electrical Engineering.

- Led development of **TRACE-CRC** for whole-trajectory uncertainty calibration in multi-step CSI prediction, obtaining tighter uncertainty regions than conservative multi-step baselines [C7].
- Led design and evaluation of **Conformal QoT**, translating calibrated predictions into lightpath-feasibility decisions, improving accuracy from 92% to 99.6% on evaluated open datasets [C4].
- Led development and expert evaluation of LLM-based explanations incorporating SHAP feature interactions; expert evaluation reported 12.2% greater usefulness than a SHAP-only prompt baseline and 97.5% correctness [C3].

AI Researcher Nov 2023 – Jul 2024
STMicroelectronics Milan, Italy

- Designed an end-to-end pipeline for continuous self-calibration of MEMS inertial sensors, integrating tiny neural networks with intelligent sensor processing units.
- Applied quantization-aware training to memory-efficient neural networks for sensor self-calibration on resource-constrained hardware [J1], [C2].

Computer Vision Intern Oct – Nov 2023
Envision The Hague, Netherlands

- Benchmarked multimodal models trained on egocentric datasets across multiple evaluation metrics.

Machine Learning Researcher Apr 2022 – Jul 2023
PoliMi Data Science Association (PMDS) Milan, Italy

- Collaborated with Sares Miramondi on production-line anomaly detection; analyzed public sentiment toward COVID-19 in a separate sentiment-analysis project.

Publications

Journal Articles

[J1] Inertial Measurement Unit Self-Calibration by Quantization-Aware and Memory-Parsimonious Neural Networks
Cardoni, M., Pau, D. P., **Rezaei, K.**, & Mura, C.
Electronics, 13(21), 4278, 2024. DOI

[J2] Conformal Risk Control for QoT-Aware Lightpath Feasibility Under Technology-Driven Domain Shift
Rezaei, K., Ayoub, O., Monti, P., & Natalino, C.
Submitted to *Journal of Lightwave Technology (JLT)*.

[J3] Reliable Pilot Scheduling for Multi-Step CSI Prediction with Calibrated Epistemic Uncertainty
Rezaei, K., Sattari, M., Aliakbari, J., Svensson, T., Monti, P., & Natalino, C.
Submitted to *IEEE Wireless Communications Letters (WCL)*.

Conference and Workshop Papers

[C7] TRACE-CRC: Trajectory-Adaptive Conformal Risk Control for Multi-Step Channel State Information Prediction
Rezaei, K., Sattari, M., Aliakbari, J., Svensson, T., Monti, P., & Natalino, C.
15th Symposium on Conformal and Probabilistic Prediction with Applications (COPA), PMLR 329, pp. 984–1009, 2026. Paper | Code

[C6] Human-Grounded Evaluation of Large Language Models for Optical Network Automation
Rezaei, K., Ayoub, O., Monti, P., & Natalino, C.
26th International Conference on Transparent Optical Networks (ICTON), pp. 1–6, 2026. Invited paper. DOI

[C5] Bridging the Trust Gap in AI-Driven Optical Networks with Structured Explainability
Rezaei, K., Ayoub, O., Natalino, C., & Monti, P.
Opto-Electronics and Communications Conference (OECC), 2026. Invited paper. Record

[C4] Policy-driven Conformal Prediction for Trustworthy QoT Estimation
Rezaei, K., Ayoub, O., Monti, P., & Natalino, C.
Optical Fiber Communication Conference (OFC), 2026. DOI

[C3] Generative Explainability for Next-Generation Networks: LLM-Augmented XAI with Mutual Feature Interactions
Rezaei, K., Ayoub, O., Troia, S., Lelli, F., Monti, P., & Natalino, C.
GenXNet Workshop, IEEE International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), 2025. DOI

[C2] IMU User Transparent Tiny Neural Self-Calibration
Cardoni, M., Pau, D. P., & **Rezaei, K.**
IEEE Forum on Research and Technologies for Society and Industry Innovation (RTSI), pp. 619–624, 2024. DOI

[C1] An Introduction to Convolutional Neural Networks & Applications
Rezaei, K. & Zamani, S.
National Conference on Contemporary Issues in Computer and Information Sciences (CICIS), 2019.

Academic Service

Peer Reviewer, *IEEE Networking Letters* 2026
Completed one manuscript review.

Executive Staff, CICIS 2019 2019
National Conference on Contemporary Issues in Computer and Information Sciences.

Teaching and Supervision

Bachelor's Thesis Supervisor 2026 (completed)
Chalmers University of Technology Gothenburg, Sweden

- Supervised the EENX16 thesis *Autonomous Knowledge Agent: From Query to Report without Human Guidance*, on autonomous AI systems.

Teaching Assistant Aug 2025 – Present
Chalmers University of Technology Gothenburg, Sweden
EEN060/EEN065: Applied Object-Oriented Programming.

- Design examinations and lead Python exercise and laboratory sessions on object-oriented programming and practical software implementation.

Industry Experience

Generative AI Engineer – Full-time Oct 2024 – Jan 2025
AGap2 / RINA Milan, Italy

- Developed LLM-based systems to assess the completeness and coherence of inspection reports.
- Integrated models into a digital reporting platform and delivered a functional minimum viable product within three months.

Data Scientist – Part-time Feb 2023 – Feb 2024
DataLobster Paris, France

- Optimized signal-processing and machine-learning algorithms for streaming data to improve performance and reduce latency.
- Developed anomaly-detection models with explainable AI to support root-cause analysis.

Selected Coursework

Conformal Prediction | KTH Royal Institute of Technology (in progress) 2026

Statistics and Machine Learning in High Dimensions | Chalmers University of Technology 2025

Advanced Digital Signal Processing | Politecnico di Milano 2023

Technical Presentation

[T1] Continuous MEMS Self-Calibration Process by Means of Tiny Neural Networks
Cardoni, M., Pau, D. P., & **Rezaei, K.**
STMicroelectronics TechWeek, 2024. Also submitted as an innovation proposal.

Professional Membership

IEEE Student Member 2025 – Present

Languages

Persian (native); English (fluent); Italian (elementary).